

AIMS System - Operation Design Interface Documentation

1. Design Class: <<interface>> IPayment

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	payOrder(amount : double, orderInfo : String, orderId : int, paymentConfig : Config)	PaymentTransaction	Initiates the payment process by creating a transaction request using the order details and amount.
2	processResponse(vnpReturnURL : String)	void	Handles the payment response from VNPay, processes transaction results and updates payment status

Parameter:

- payOrder(amount: double, orderInfo: String): PaymentTransaction
 - **amount:** The amount to be paid.
 - **orderInfo:** Detailed information about the order.
 - **orderId:** Unique identifier for the order within the AIMS system.
 - **paymentConfig:** Configuration object containing necessary parameters for VNPay integration, including vnpVersion, vnpTmnCode, vnpReturnUrl, vnpSecureHash, and other connection settings required for secure payment processing.
- processResponse(vnpReturnURL: String): void
 - **vnpReturnURL:** The URL returned from VNPay containing payment response data including transaction status and details.

Exception:

- payOrder(amount: double, orderInfo: String): void
 - **NotEnoughBalanceException:** This exception is thrown when the user’s bank account or linked payment method does not have sufficient funds to complete the transaction.
 - **TimeoutException:** This exception is triggered when the payment request or response takes too long to process, exceeding the predefined time limit.
 - **SuspiciousException:** This exception indicates that the transaction is flagged as potentially fraudulent or suspicious based on predefined security rules.
 - **UnregisteredInternetBankingException:** This exception is thrown when the user tries to make an online payment using a bank account that is not registered for internet banking services.
- processResponse(vnpReturnURL: String): void
 - **PaymentException:** Thrown when the response indicates payment failure or contains invalid data.

Method:

How to use parameters/attributes:

- **payOrder(amount: double, orderInfo: String): void**
 - **Purpose:** This method initiates the payment process for a specific order using the provided amount and order details. It prepares a transaction request to be sent to VNPay.
 - **Usage:** Called when the user confirms an order and proceeds to payment. The method sends a request to the VNPay API with the given amount and order information. Exceptions may occur if the request fails or validation errors happen.
- **processResponse(vnpReturnURL: String): void**
 - **Purpose:** Processes the payment gateway's response by parsing the return URL and extracting transaction results.
 - **Usage:** Called when VNPay redirects back to the application after payment processing. This method validates the response, extracts transaction details, and updates the order status accordingly.

State:

- When this method is called, the system transitions from the "Order Confirmed" state to the "Payment Processing" state. A payment request is created and sent to VNPay. If the request is successful, the system moves to the "Waiting for Payment Response" state.
- If an exception occurs (e.g., insufficient balance, timeout, suspicious activity, or unregistered internet banking), the system transitions to the "Payment Failed" state, and appropriate error handling is triggered.
- When processResponse is called, the system transitions from "Waiting for Payment Response" to either "Payment Successful" or "Payment Failed" based on the response data. If successful, the order status is updated and transaction details are recorded.